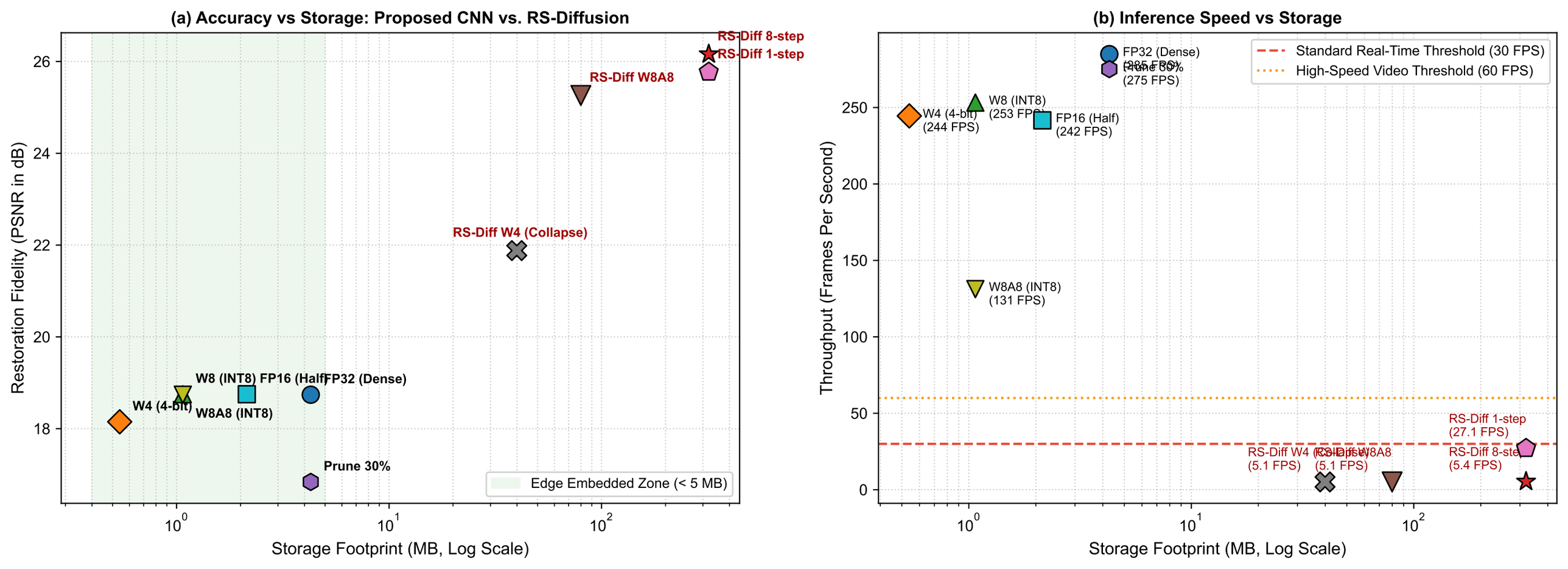
(a) Accuracy vs Storage: Proposed CNN vs. RS-Diffusion
Restoration Fidelity (PSNR in dB)
Storage Footprint (MB, Log Scale)
RS-Diff 8-step
RS-Diff 1-step
RS-Diff W8A8
RS-Diff W4 (Collapse)
W8 (INT8)
FP16 (Half)
FP32 (Dense)
W4 (4-bit)
W8A8 (INT8)
Prune 30%
Edge Embedded Zone (< 5 MB)
(b) Inference Speed vs Storage
Throughput (Frames Per Second)
Storage Footprint (MB, Log Scale)
Standard Real-Time Threshold (30 FPS)
High-Speed Video Threshold (60 FPS)
FP32 (Dense)
(275 FPS)
W8 (INT8)
(253 FPS)
W4 (4-bit)
(244 FPS)
FP16 (Half)
(242 FPS)
W8A8 (INT8)
(131 FPS)
RS-Diff 1-step
(27.1 FPS)
RS-Diff W4 (Collapse)
(5.1 FPS)
RS-Diff W8A8
(5.1 FPS)
RS-Diff 8-step
(5.4 FPS)